CURRICULUM VITAE

Endawoke Yizengaw

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I. EDUCATIONAL BACKGROUND

- **Ph.D. in space physics**, La Trobe University in Melbourne, Australia: May 2004.
- **M.Sc. in atmospheric physics**, Tromso University in Tromso, Norway: May 1998.
- **B.Sc. in applied Physics**, Addis Ababa University in Addis Ababa, Ethiopia: July 1994.

II. EXPERIENCES

- 1 July 2009 - Present, **Senior Research Scientist,** ISR, Boston College, USA.
- 1 July 2006 - 30 June 2009, **Assistant Research Geophysicist,** UCLA, IGPP, USA
- May 2004 – June 2006, **Postdoctoral Researcher,** UCLA, IGPP, USA

III. SOME OF SUCCESSFUL GRANTS

1. **Principal-Investigator**, "African Meridian B-field Education and Research (AMBER) Magnetometer Network", 2008 – Present. (**Funded by NASA, AFOSR, and NSF**)
2. **Principal-Investigator**, "South American Meridian B-field Array (SAMBA) Magnetometer Network", 2013 – Present. (**Funded by NSF and AFOSR**)
3. **Co-Principal-Investigator**, NSF OPP, Magnetosphere, and Aeronomy programs, "iMAGS: inner-Magnetosphere Array for Geospace Science", 2015-2018.
4. **Principal-Investigator**, AFOSR Space Science program, "Investigation of the longitudinal dependence of ionospheric density distribution and its driving mechanisms", 2015-2018.
5. **Co-Principal-Investigator**, AFOSR Space Science program, "Specifying ionospheric and plasmaspheric dynamics using ground-based magnetometers", 2012-2015.
6. **Principal-Investigator**, AFOSR Space Science program, "Investigation of the longitudinal dependence of ionospheric density distribution and its driving mechanisms", 2012-2015.
7. **Principal-Investigator**, NSF Aeronomy program, "Meeting Support for the International AGU Chapman Conference, Addis Ababa, Ethiopia, November 12-16, 2012", 2012-2013.
8. **Principal-Investigator**, NSF Facility Pprogram, "Advanced Modular Incoherent Scatter Radar in Ethiopia: Proposal for Meeting to Identify Science Rationale", 2011-2012.
9. **Principal-Investigator**, NASA LWS TRT, "Investigation of the low-latitude Electrodynamics and Seeding Conditions of Plasma Structures by Utilizing Multi-instrument Observations", 2011-2015.
10. **Co-Principal-Investigator**, AFOSR Space Science program, "Ionospheric Forecasting and Prediction with SAMBA Magnetometers", 2010-2012.
11. **Co-Principal-Investigator**, NASA LWSTRT, "A 3D climate and weather global topside ionosphere and plasmasphere model", 2010-2014.
12. **Principal-Investigator**, NASA Geospace program, "Imaging Ionospheric Outflow Events", 2009-2011.
13. **Principal-Investigator**, AFOSR-YIP FY09 program, "Understanding the physics behind ionospheric and plasmaspheric density irregularities by utilizing multi-instrument observations data", 2010-2013.
14. **Principal-Investigator**, NSF Aeronomy program, "Meeting Support for the International Heliophysical Year (IHY) Africa 2009 Workshop Livingstone, Zambia, June 8-12, 2009, 2009 – 2010.
15. **Principal-Investigator**, NASA IHY grant, "African Meridian B-field Education and Research (AMBER) Array", 2007 – 2010.
16. **Co-Principal-Investigator**, NASA Heliophysics Guest Investigator program, "Plasmaspheric Density response to Geomagnetic Storms", 2008-2011.

17. **Principal-Investigator**, NSF GEM/CEDAR postdoc, "Understanding Magnetospheric-Ionospheric coupling through tomographic imaging of the ionospheric and plasmaspheric density using ground- and space-based GPS receivers", 2005-2007.

IV. HONORS RECEIVED

- **Fellow of American Geophysical Union** (AGU) (2018 – Present)
- **The 2018 AGU's Joanne Simpson Medal for Mid-Career Scientists** for his significant contribution to space sciences
- **Fellow of African Geophysical Society** (2014 – Present)
- **The 2011 ISR-Boston College Annual award** for Scientific Excellence in serving as Principal Investigator for grant agreements with NSF, NASA, and AFOSR
- **The 2010 ISR-Boston College Annual award** for outstanding performance in International outreach program that expand space weather research
- **The 2009 UN-IHY award** for best creative vision & outstanding leadership
- **American Geophysical Union (AGU) *Certificate of Appreciation*** for his outstanding Contribution to SPA-EPO.
- Won the **2006 *UCLA Chancellor's Award* for best Postdoctoral Researchers**
- Won the **best student presentation prize at the Australian national URSI conference**

V. PROFESSIONAL COMMUNITY SERVICES

- Member of the national ground magnetometer array advisory board to NSF (05/2017 –present)
- Chair, committee of AGU's Africa Awards for Research Excellence in Earth and Space Science (01/2016 – Present)
- Associate Editor of Radio Science Journal (01/2016 – Present)
- Editor in-chief for the Annales Geophysicae special issue of *"The 14th International Symposium on Equatorial Aeronomy"* (07/2016 – 08/2017).
- Editor, AGU book "*Ionospheric Space Weather: Longitude and Hemispheric Dependences and Lower Atmosphere Forcing*"
- Main scientific organizing committee of the 2018 ISEA-15 Symposium, Ahmedabad, India
- Chaired the main scientific organizing committee of the 2015 ISEA-14 Symposium, Bahir Dar, Ethiopia
- Served as proposal review panelist for NASA, NSF, AFOSR
- Co-convened the 2012 International AGU's Chapman Conference in Africa
- Co-convened the 2010, 2011, 2017 & 2018 Fall AGU SPA session
- Co-organize ISWI 2010 Summer School, Bahir Dar, Ethiopia
- Co-organize IHY-Africa 2009 meeting, Livingstone, Zambia
- Co-organize GIFT workshop, Livingstone, Zambia
- Co-organize IHY-Africa 2007 meeting, Addis Ababa, Ethiopia
- Co-organize GIFT workshop, Addis Ababa, Ethiopia

VI. PROFESSIONAL MEMBERSHIP

- Member of American Geophysical Union (AGU).
- Associate member of Committee on Space Research (COSPAR)
- Members of the Australian Institute of Physics (AIP).
- Founding Member of Ethiopian Geophysical Union International (EGUI)

VII. PUBLICATIONS & PRESENTATIONS

A. BOOK EDITOR

1. Ionospheric Space Weather: Longitude Dependence and Lower Atmosphere Forcing (2016), Timothy Fuller-Rowell, **Endawoke Yizengaw**, Patricia H. Doherty, Sunanda Basu (Editors), *American Geophysical Union, ISBN: 978-1-118-92920-9*.

CURRICULUM VITAE

B. JOURNAL PUBLICATIONS

1. **Yizengaw, E.** (2018), Ionospheric Dynamics and Its Strong Longitudinal Dependences, *In C-S. Huang and G. Lu (Eds.), Advances in Ionospheric Research: Current Understanding and Challenges*, *American Geophysical Union (Under Review)*
2. Ngwira,C. M., J. B. Habarulema, E. Astafyeva, **E. Yizengaw**, G. Crowley, A. Gisler, and V. Coffey (2018), Dynamic response of ionospheric electron plasma density during the geomagnetic storm of 22-23 June 2015, *J. Geophys. Res. 122 (Under Review)*
3. Scherliess, L., I. Tsagouri, **E. Yizengaw**, S. Bruinsma, J-S. Shim, A. Coster, and J. Retterer (2018), The iCCMC Ionosphere/Thermosphere Modeling Capabilities Assessment: Overview and Initial Results, *Space Weather. 14, (Under Review)*
4. Habarulema, J. B., M. B. Dubazane, Z. T. Katamzi-Joseph, **E. Yizengaw**, M. B. Moldwin, J. C. Uwamahoro (2018), Long-term estimation of diurnal vertical E × B drift velocities using C/NOFS and ground-based magnetometer observations, *J. Geophys. Res. 122, doi:10.1029/2018JA025685.*
5. Pradipta, R., **E. Yizengaw**, and P. H. Doherty (2018), Ionospheric Density Irregularities, Turbulence, and Wave Disturbances during the Total Solar Eclipse over North America on 21 August 2017, *Geophys. Res. Lett. 45, doi:10.1029/2018GL079383.*
6. **Yizengaw, E.** and K. Groves (2018), Longitudinal Variability of Equatorial Electrodynamics, Ionospheric Irregularities, and Ambient Electron Density, *Space Weather. 14, doi:10.1029/2018SW001980.*
7. **Yizengaw, E.**, E. Zesta, M. B. Moldwin, M. Magoun, N. K. Tripathi, C. Surussavadee, and Z. Bamba (2018), ULF Wave Associated Density Irregularities and Scintillation at the Equator, *Geophys. Res. Lett. 45, doi:10.1029/2018GL078163.*
8. Oliveira, D. M., D. Arel, J. Raeder, E. Zesta, C. M. Ngwira, B. A. Carter, **E. Yizengaw**, A. J. Halford, B. T. Tsurutani, and J. W. Gjerloev (2018), Geomagnetically induced currents caused by interplanetary shocks with different impact angles and speeds, *Space Weather. 14, doi:10.1029/2018SW001880.*
9. Tebabal, A., S.M. Radicella, M. Nigussie, B. Damtie, B. Nava, **E. Yizengaw** (2018), Local TEC modelling and forecasting using neural networks, *J. Atmos. Solar-Terr. Phys., 172, 143-151, doi:10.1016/j.jastp.2018.03.004.*
10. Carter, B., S. T. Ram, **E. Yizengaw**, R. Pradipta, J. Retterer, R. Norman, J. Currie, K. Groves, R. Caton, M. Terkildsen, T. Yokoyama, and K. Zhang (2018), Unseasonal development of F-region irregularities over Southeast Asia on 28 July 2014: Forcing from above?, *Prog. Earth and Planet. Sci. 5:10, DOI 10.1186/s40645-018-0164-y.*
11. Habarulema, J. B., **E. Yizengaw**, Z. T. Katamzi-Joseph, M. B. Moldwin, S. Buchert (2018), Storm time global observations of large-scale TIDs from ground-based and in situ satellite measurements, *J. Geophys. Res. 122, doi:10.1002/2017JA024510.*
12. Tebabal, A., B. Damtie, M. Nigussie, and **E. Yizengaw** (2017), Temporal Variations in Solar Irradiance Since 1947, *Sol Phys., 292: 112, doi.org/10.1007/s11207-017-1128-x.*
13. **Yizengaw, E.** and B. Carter (2017), Longitudinal, Seasonal and Solar Cycle Variation of Lunar Tide Influence on the Equatorial Electrojet, *Ann. Geophys., 35, 525-533, doi:10.5194/angeo-35-525-2017.*
14. Mazzella, A., J. Habarulema, and **E. Yizengaw** (2017), Determinations of ionosphere and plasmasphere electron content for an African chain of GPS stations, *Ann. Geophys., 35, doi:10.5194/angeo-35-599-2017.*
15. Tesema, T., R. Mesquita, J. Meriwether, B. Damtie, M. Nigussie, J. Makela, D. Fisher, B. Harding, **E. Yizengaw**, and S. Sanders (2017), New results on equatorial thermospheric winds and temperatures from Ethiopia, Africa, *Ann. Geophys., 35, 333–344, doi:10.5194/angeo-35-333-2017.*
16. Bolaji, O., O. Owolabi, E. Falayi, E. Jimoh, A. Kotoye, O. Odeyemi, B. Rabiu, P. Doherty, **E. Yizengaw**, Y. Yamazaki, J. Adeniyi, R. Kaka, and K. Onanuga (2017), Observations of equatorial ionization anomaly over Africa and Middle East during a year of deep minimum, *Ann. Geophys., 35, 123–132, doi/0.5194/angeo-35-123-2017.*
17. Carter, B. A., **E. Yizengaw**, E., R. Pradipta, J. Weygand, A. Pulkkinen, M. Moldwin, M. Piersanti, R. Norman, and K. Zhang (2016), Geomagnetically induced currents around the world during the 2015 St. Patrick's Day storm, *J. Geophys. Res. 121, doi/10.1002/2016JA023344.*

Post-doc he supervised

PhD student he supervised

18. **Yizengaw, E.**, M. B. Moldwin, E. Zesta, M. Magoun, R. Pradipta, C. M. Biouele, A. B. Rabiu, O. K. Obrou, Z. Bamba, and E. R. de Paula (2016), Response of the Equatorial Ionosphere to the Geomagnetic DP2 current system, *Geophys. Res. Lett. 43, doi:10.1002/2016GL070090.*
19. Habarulema, J. B., Z. T. Katamzi, **E. Yizengaw**, Y. Yamazaki, and G. Seemala (2016), Simultaneous storm time equatorward and poleward largescale TIDs on a global scale, *Geophys. Res. Lett. 43, doi:10.1002/2016GL069740.*
20. Nigussie M., S. M. Radicella, B. Damtie, **E. Yizengaw**, B. Nava, and L. Roininen (2016), Validation of NeQuick TEC data ingestion technique against C/NOFS and EISCAT electron density measurements, *Radio Sci., 51, doi:10.1002/2015RS005930.*
21. Anad, A., C. Amory-Mazaudier, M. Hamoudi, S. Bourouis, A. Abtout, and **E. Yizengaw** (2016), Sq solar variation at Medea Observatory (Algeria), from 2008 to 2011, *Adv. Space Res., doi:10.1016/j.asr.2016.06.029.*
22. **Yizengaw, E.** and B. A. Carter (2016), Imaging the global vertical density structure from the ground and space, *Timothy Fuller-Rowell, Endawoke Yizengaw, Patricia H. Doherty, Sunanda Basu (Editors), (2016), American Geophysical Union, ISBN: 978-1-118-92920-9.*
23. Zesta, E., A. Boudouridis, J.M. Weygand, **E. Yizengaw**, M. B. Moldwin, and P. Chi (2016), Inter-Hemispheric Asymmetries in Magnetospheric Energy Input, *Timothy Fuller-Rowell, Endawoke Yizengaw, Patricia H. Doherty, Sunanda Basu (Editors), (2016), AGU, ISBN: 978-1-118-92920-9.*
24. Nigussie, M., B. Damtie, **E. Yizengaw**, and S. M. Radicella (2016), Modeling the East-African Ionosphere, *Timothy Fuller-Rowell, Endawoke Yizengaw, Patricia H. Doherty, Sunanda Basu (Editors), (2016), American Geophysical Union, ISBN: 978-1-118-92920-9.*
25. Carter, B. A., **E. Yizengaw**, E., R. Pradipta, J. Retterer, K. Groves, C. Valladares, R. Caton, C. Bridgwood, R. Norman, and K. Zhang (2016), Global equatorial plasma bubble occurrence during the 2015 St. Patrick's Dat Storm, *J. Geophys. Res. 121, 894–905, doi:10.1002/2015JA022194.*
26. Carter, B. A., **E. Yizengaw**, R. Pradipta, A. J. Halford, R. Norman, K. Zhang (2015), Interplanetary shocks and the resulting geomagnetically induced currents at the equator, *Geophys. Res. Lett., 42, 6554–6559, doi:10.1002/2015GL065060.*
27. Habarulema, J. B., Z. Katamzi, and **E. Yizengaw** (2015), First observations of poleward large scale travelling ionospheric disturbances over the African sector during the geomagnetic storm conditions, *J. Geophys. Res., 120, 6914–6929, doi:10.1002/2015JA021066.*
28. Maute, A, M.E. Hagan, V. Yudin, H.-L. Liu, and **E. Yizengaw** (2015), Causes of the longitudinal differences in the equatorial vertical ExB drift during the 2013 SSW period as simulated by the TIME-GCM, *J. Geophys. Res. Space Physics 120, 5117–5136. doi: 10.1002/2015JA021126.*
29. Tebabal, A., B. Damtie, M. Nigussie, A. Bires, and **E. Yizengaw** (2015), Modeling total solar irradiance from PMOD composite using feed-forward neural networks, *J. Atmos. Solar-Terr. Phys., 135, 64-71.*
30. Kassa, T., B. Damtie, A. Bires, **E. Yizengaw**, and P. Cilliers (2015), Storm-time characteristics of the equatorial ionization anomaly in the East African sector, *Adv. Space Res., 56(1), 57-70.*
31. Kassa, T., B. Damtie, A. Bires, **E. Yizengaw**, and P. Cilliers (2015), Spatio-temporal characteristics of the Equatorial Ionization Anomaly (EIA) in the East African region via ionospheric tomography during the year 2012, *Adv. Space Res., 55(1), 184-198.*
32. Carter, B. A., J. M. Retterer, **E. Yizengaw**, K. Wiens, S. Wing, K. Groves, R. Caton, C. Bridgwood, M. Francis, M. Terkildsen, R. Norman, and K. Zhang (2014), Using solar wind data to predict daily GPS scintillation occurrence in the African and Asian low-latitude regions, *Geophys. Res. Lett., 41, 8176–8184, doi:10.1002/2014GL062203*.
33. Habarulema, J. B., Z. T. Katamzi, and **E. Yizengaw** (2014), A simultaneous study of ionospheric parameters derived from FORMOSAT-3/COSMIC, GRACE and CHAMP missions over mid, low and equatorial latitudes: comparison with ionosonde data, *J. Geophys. Res. Space Physics, 119, doi:10.1002/2014JA020192*.

PhD student he supervised
Post-doc he supervised

34. Carter, B. A., J. M. Retterer, **E. Yizengaw**, K. Groves, R. Caton, L. McNamara, C. Bridgwood, M. Francis, M. Terkildsen, R. Norman, and K. Zhang (2014), Geomagnetic control of equatorial plasma bubble activity modeled by the TIEGCM with Kp, *Geophys. Res. Lett., 41, doi:10.1002/2014GL060953*.
35. **Yizengaw, E.**, M. B. Moldwin, E. Zesta, C. M. Biouele, B. Damtie, A. Mebrahtu, B. Rabiu, C. E. Valladares, and R. Stoneback (2014), The longitudinal variability of equatorial electrojet and vertical drift velocity in the African and American sectors, *Ann. Geophys., 32, 231–238, 2014*.
36. Carter, B. A., **E. Yizengaw**, J. M. Retterer, M. Francis, M. Terkildsen, R. Marshall, R. Norman, and K. Zhang (2014), An analysis of the quiet time day-to-day variability in the formation of postsunset equatorial plasma bubbles in the Southeast Asian region, *J. Geophys. Res. Space Physics, 119, doi:10.1002/2013JA01957*.
37. Bello, O.R., A.B. Rabiu, K. Yumoto, and **E. Yizengaw** (2014), Mean solar quiet daily variations in the earth's magnetic field along East African longitudes, *J. Adv. Space Res., 54(3), 283-289I.*
38. **Yizengaw, E.**, J. Retterer, E. E. Pacheco, P. Roddy, K. Groves, R. Caton, and P. Baki (2013), Post-midnight Bubbles and Scintillations in the Quiet-Time June Solstice, *Geophys. Res. Lett., 40, doi:10.1002/2013GL058307*.
39. **Yizengaw, E.**, E. Zesta, C. M. Biouele, M. B. Moldwin, A. Boudouridis, B. Damtie, A. Mebrahtu, F. Anad, R. F. Pfaff, and M. Hartinger (2013), Observations of ULF wave related equatorial electrojet fluctuations, *J. Atmos. Solar-Terr. Phys., 103, 157-168.*
40. **Yizengaw, E.**, P. Doherty, and T. Fuller-Rowell (2013), Is Space Weather Different Over Africa, and If So, Why? AGU Chapman Conference Report, *Space Weather, 11, doi:10.1002/swe.20063.*
41. Pacheco, E. E. and **E. Yizengaw** (2013), The day-to-day longitudinal variability of the global ionospheric density distribution at low latitudes during low solar activity, *J. Geophys. Res. Space Physics, 118, doi:10.1002/jgra.50241*,
42. Fuller-Rowell, T., **E. Yizengaw**, P. Doherty, and G. Mengistu (2013), The Longitudinal Properties of the Ionosphere and Their Effects on Space Weather, *Eos, Transactions AGU, 94(17), 23 April 2013, pp: 160.*
43. Nigussie, M., S.M. Radicella, B. Damtie, B. Nava, **E. Yizengaw**, and K. Groves (2013), Validation of the NeQuick 2 and IRI-2007 models in East-African equatorial region, *J. Atmos. Solar-Terr. Phys., 102, pp:26*.
44. Zou, S., M. B. Moldwin, M. J. Nicolls, A. J. Ridley, A. J. Coster, **E. Yizengaw**, L. R. Lyons, and E. F. Donovan (2013), Electrodynamics of the high-latitude trough: its relationship with convection flows and field-aligned currents, *J. Geophys. Res. Space Physics, 118, 2565–2572, doi:10.1002/jgra.50120.*
45. Nigussie, M., S.M. Radicella, B. Damtie, B. Nava, **E. Yizengaw**, and L. Ciraolo (2012),TEC assimilation to the NeQuick 2 for modeling the East-African equatorial region ionosphere, *Radio Sci., 47, RS5002, doi:10.1029/2012RS004981.*
46. **Yizengaw, E.**, E. Zesta, M. B. Moldwin, B. Damtie, A. Mebrahtu, C. E. Valladares, and R. F. Pfaff (2012), Longitudinal differences of ionospheric vertical density distribution and equatorial electrodynamics, *J. Geophys. Res.,117, A07312, doi:10.1029/2011JA017454.*
47. **Yizengaw, E.** (2012), Global Longitudinal Dependence Observation of the Neutral Wind and Ionospheric Density Distribution, *Int. J. Geophys., vol. 2012, doi:10.1155/2012/342581*, **(Invited Paper)**
48. Norman, R. J., P. L. Dyson, **E. Yizengaw**, J. Le Marshall, C. Wang, B. Carter, D. Wen and K. Zhang (2012), Radio Occultation Measurements from the Australian Micro Satellite FedSat, *IEEE Trans. Geo. Remote Sens., 10.1109/TGRS.2012.2194295.*
49. Ngwira, C. M., L. McKinnell, P. J. Cilliers, **E. Yizengaw**, and B. D. L. Opperman (2012), An investigation of ionospheric disturbances over South Africa during the magnetic storm on 15 May 2005, *J. Adv. Space Res., 49(2), 327-335.*
50. **Yizengaw, E.**, M. B. Moldwin, A. Mebrahtu, B. Damtie, E. Zesta, C. E. Valladares, and P. H. Doherty (2011), Comparison of storm time equatorial ionospheric electrodynamics in the African and American sectors, *J. Atmos. Solar-Terr. Phys., 73(1), 156-163*.
51. **Yizengaw, E.**, M. B. Moldwin, Y. Sahai, and R. de Jesus (2009), Strong Post-Midnight Equatorial Ionospheric Anomaly Observations during Magnetically Quiet Periods, *J. Geophys. Res., 114, A12308, doi:10.1029/2009JA014603.*

Post-doc he supervised

PhD student he supervised

52. **Yizengaw, E.**, and M. B. Moldwin (2009), African Meridian B-field Education and Research (AMBER) Array, *Earth Moon Planet,104(1), 237-246, doi:10.1007/s11038-008-9287-2.*
53. Amory-Mazaudier, C., S. Basu, O. Bock, A. Combrink, K. Groves, T. Fuller Rowell, P. Lassudrie-Duchesne, M. Petitdidier, and **E. Yizengaw** (2009), The use of Global Positioning System to derive atmospheric water vapor distribution for environmental applications, *Earth Moon Planet,104(1), 263-270, doi:10.1007/s11038-008-9273-8.*
54. **Yizengaw, E.**, M. B. Moldwin, D. Galvan, B. A. Iijima, A. Komjathy, and A. Mannucci (2008), The global plasmaspheric TEC and its contribution to the GPS TEC, *J. Atmos. Solar-Terr. Phys., 70, 1541-1548.*
55. **Yizengaw, E.**, J. Dewar, J. MacNeil, M. B. Moldwin, D. Galvan, J. Sanny, D. Berube, and Bill Sandel (2008), The occurrence of Ionospheric Signatures of Plasmaspheric Plumes over Different Longitudinal Sectors, *J. Geophys. Res., 113, A08318, doi:10.1029/2007JA012925.*
56. Moldwin, M. B., **E. Yizengaw**, D. K. Scherrer, M. C. Rabellosoares, and P. Reiff (2008), AGU Scientists Host Teacher Workshop in Ethiopia, *Eos, Vol. 89, No. 10, 4 March 2008. Documentary movie was also produced by Geoffrey Haines- Stiles and NASA's Passport to Knowledge team*.
57. **Yizengaw, E.**, M. B. Moldwin, P. L. Dyson, E. A. Essex (2007), Using Tomography of GPS TEC to Routinely Determine Ionospheric Average Electron Density Profiles, *J. Atmos. Solar-Terr. Phys., 69, 314–321.*
58. **Yizengaw, E.**, P. L. Dyson, and E. A. Essex (2006), Tomographic observations of the topside ionospheric density distributions using the GPS receiver on FedSat, *Adv. Space Res., 38, 2318-2323.*
59. **Yizengaw, E.**, M. B. Moldwin, P. L. Dyson, B. J. Fraser, and S. K. Morley (2006), First Tomographic Images of Ionospheric Outflows, *Geophys. Res. Lett., 33, L20102, doi:10.1029/2006GL027698*. (***Featured on the journal's cover page***).
60. **Yizengaw, E.**, M. B. Moldwin, and D. Galvan (2006), Ionospheric signatures of a plasmaspheric plume over Europe, *Geophys. Res. Lett., 33, L17103, doi:10.1029/2006GL026597.*
61. **Yizengaw, E.**, M. B. Moldwin, A. Komjathy, and A. J. Mannucci (2006), Unusual topside ionospheric density response to the November 2003 superstorm, *J. Geophys. Res., 111, A02308, doi:10.1029/2005JA011433*.
62. **Yizengaw, E.**, H. Wei, M. B. Moldwin, D. Galvan, L. L. Mandrake, A. Mannucci, and X. Pi (2005), The correlation between mid-latitude trough and the plasmapause, *Geophys. Res. Lett., 32, L10102, doi:10.1029/2005GL022954*.
63. **Yizengaw, E.**, and M. B. Moldwin (2005), The altitude extension of the mid-latitude trough and its correlation with plasmapause position, *Geophys. Res. Lett.,32, L09105, doi:10.1029/2005GL022854*. (***Featured on the journal's cover page***).
64. **Yizengaw, E.**, M. Moldwin, P. L. Dyson, and T. J. Immel (2005), The Southern Hemisphere ionosphere and plasmasphere response to the interplanetary shock event of 29-31 October 2003, *J. Geophys. Res.,110, A09S30, doi:10.1029/2004JA010920*.
65. **Yizengaw, E.**, P. L. Dyson, E. A. Essex, and M. B. Moldwin (2005), Ionosphere dynamics over the Southern Hemisphere during the 31 March 2001 severe magnetic storm using multi-instrument measurement data, *Ann. Geophys. 23(3), 707-721*.
66. **Yizengaw, E.**, E. A. Essex, and R. Birsa (2004), The Southern Hemisphere and Equatorial region ionization response for September 22, 1999 severe magnetic storm, *Ann. Geophys. 22(8), 2765-2773*.
67. Parkinson, M. L., P. L. Dyson, M. Pinnock, J. C. Devlin, M. R. Hairston, **E. Yizengaw**, and P. J. Wilkinson (2002), Signatures of the midnight open-closed magnetic field-line boundary during balanced dayside and nightside reconnection, *Ann. Geophys., 20(10), 1617-1630*.
68. **Yizengaw, E.**, C.M. Hall, A.H. Manson, & C.E. Meek (1999), Nonvalidation of mesospheric horizontal wind fluctuations derived from temperature data via comparison with MF radar wind measurements, *J. Geophys. Res. 104(D22), 27565-27572*.
69. **Yizengaw, E.**, P.L. Dyson, and E.A. Essex, Tomographic reconstruction of the ionosphere using ground-based GPS data in the Australian region, *Workshop on Applications of Radio Science (WARS), Hobart, Australia, February* 2004.
70. **Yizengaw, E.**, and E.A. Essex, Storm Time Seasonal Variation of TEC on the Southern Hemisphere Mid-Latitude Regions Using Signals from Gps Satellite, *OIST-4 proceedings, 221-224*, 2002.

71. **Yizengaw, E.**, and E.A. Essex, Use of GPS signals to study total electron content of the ionosphere during the geomagnetic storm on 22 September 199, *Workshop on Applications of Radio Science (WARS), Blue Mountains, Australia, February* 2002.